

NAME OF THE MEDICINAL PRODUCT

Ziromin 200 mg/5 mL Powder for Oral Suspension

QUALITATIVE AND QUANTITATIVE COMPOSITION**Active substance:**

In Each 5 ml: Azithromycin dihydrate 209.6 mg (equivalent to 200.00 mg azithromycin)

Product Description

Powder: White colored, cherry scented, homogeneous powder free from particles.

Suspension: White or off-white homogeneous suspension.

Pharmacodynamics

Azithromycin is a macrolide antibiotic belonging to the azalide group. The molecule is constructed by adding a nitrogen atom to the lactone ring of erythromycin A. The mechanism of action of azithromycin is based upon the suppression of bacterial protein synthesis by means of binding to the ribosomal 50S sub-unit and inhibition of peptide translocation.

Resistance to azithromycin may be inherent or acquired. There are three main mechanisms of resistance in bacteria: target site alteration, alteration in antibiotic transport and modification of the antibiotic.

Azithromycin demonstrates cross resistance with erythromycin resistant gram positive isolates. A decrease in macrolide susceptibility over time has been noted particularly in *Streptococcus pneumoniae* and *Staphylococcus aureus*. Similarly, decreased susceptibility has been observed among *Streptococcus viridans* and *Streptococcus agalactiae* (Group B) streptococcus against other macrolides and lincosamides.

Antibacterial spectrum of Azithromycin:

Commonly susceptible species:

Aerobic Gram-positive microorganisms: *Staphylococcus aureus*, Methicillin-susceptible, *Streptococcus pneumoniae*, Penicillin-susceptible, *Streptococcus pyogenes* (Group A).

Aerobic Gram-negative microorganisms: *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Legionella pneumophila*, *Moraxella catarrhalis*, *Neisseria gonorrhoeae*, *Pasteurella multocida*

Anaerobic microorganisms: *Clostridium perfringens*, *Fusobacterium* spp., *Prevotella* spp., *Porphyromonas* spp.

Other microorganisms: *Chlamydia trachomatis*.

Species for which acquired resistance:

Aerobic Gram-positive microorganisms: *Streptococcus pneumoniae*, Penicillin-intermediate Penicillin-resistant.

Inherently resistant organisms:

Aerobic Gram-positive microorganisms: *Enterococcus faecalis*, *Staphylococci* MRSA, MRSE*

Anaerobic microorganisms: *Bacteroides fragilis* group

* Methicillin-resistant staphylococci have a very high prevalence of acquired resistance to macrolides and have been placed here because they are rarely susceptible to azithromycin.

Pharmacokinetics

Absorption

Following oral administration in humans, azithromycin is widely distributed throughout the body; bioavailability is approximately 37%. The time taken to peak plasma levels is 2-3 hours.

Distribution

Orally administered azithromycin is widely distributed throughout the body. It has been demonstrated that the concentrations of azithromycin are markedly higher levels in tissue than in plasma (up to 50 times the maximum observed concentration in plasma) indicating that the drug is heavily tissue bound. Concentrations in target tissues, such as lung, tonsil and prostate exceed the MIC₉₀ for likely pathogens after a single dose of 500 mg.

Elimination

Plasma terminal elimination half-life closely reflects the tissue depletion half-life of 2 to 4 days. Approximately 12% of an intravenously administered dose is excreted in the urine over 3 days as the parent drug, the majority in the first 24 hours. Biliary excretion of azithromycin is a major route of elimination for unchanged drug following oral administration. Very high concentrations of unchanged drug have been found in human bile, together with 10 metabolites, formed by N- and O-demethylation, by hydroxylation of the desosamine and aglycone rings, and by cleavage of the cladinose conjugate. Comparison of HPLC and microbiological assays in tissues suggests that metabolites play no part in the microbiological activity of azithromycin.

Pharmacokinetics in Special Patient Groups

Elderly

In elderly patients (>65 years), slightly higher AUC values were seen after a 5-day regimen than in young patients (<40 years), but these are not considered clinically significant, and hence no dose adjustment is recommended.

Renal Impairment

The pharmacokinetics of azithromycin in individual with mild to moderate renal impairment (GFR 10 - 80 ml/min) were not affected following a single one-gram dose of immediate release azithromycin. Statistically significant differences in AUC 0-120 (8.8 µg·hr/ml vs. 11.7 µg·hr/ml), C_{max} (1.0 µg/ml vs. 1.6 µg/ml) and CL_r (2.3 ml/min/kg vs. 0.2 ml/min/kg) were observed between the group with severe renal impairment (GFR < 10 ml/min) and the group with normal renal function.

Hepatic Impairment

In patients with mild (Class A) to moderate (Class B) hepatic impairment, there is no evidence of a marked change in serum pharmacokinetics of azithromycin compared to those with normal hepatic function. In these patients urinary clearance of azithromycin appears to increase, perhaps to compensate for reduced hepatic clearance.

Indication

Azithromycin is indicated for infections caused by susceptible organisms; in lower respiratory tract infections including bronchitis and pneumonia, in skin and soft tissue infections, in acute otitis media and in upper respiratory tract infections including sinusitis and pharyngitis/tonsillitis. (Penicillin is the usual drug of choice in the treatment of *Streptococcus pyogenes* pharyngitis, including the prophylaxis of rheumatic fever. Azithromycin is generally effective in the eradication of streptococci from the oropharynx, however, data establishing the efficacy of azithromycin and the subsequent prevention of rheumatic fever are not available at present.)

In sexually transmitted diseases in men and women, azithromycin is indicated in the treatment of uncomplicated genital infections due to *Chlamydia trachomatis*. It is also indicated in the treatment of chancroid due to *Haemophilus ducreyi*; and uncomplicated genital infection due to non-multiresistant *Neisseria gonorrhoea*; concurrent infection with *Treponema pallidum* should be excluded.

Recommended Dosage

Oral azithromycin should be administered as a single daily dose. The period of dosing with regard to infection is given below. Azithromycin powder for oral suspension can be taken with or without food.

Adults

For the treatment of sexually transmitted diseases caused by *Chlamydia trachomatis* and *Haemophilus ducreyi* or susceptible *Neisseria gonorrhoea*, the dose is 1000 mg as a single oral dose. For patients who are allergic to penicillin and/or cephalosporins, prescribers should consult local treatment guidelines.

For all other indications in which the oral formulation is administered, the total dosage of 1500 mg should be given as 500 mg daily for 3 days. As an alternative, the same total dose can be given over 5 days with 500 mg given on day 1, then 250 mg daily on days 2 to 5.

Children

The maximum recommended total dose for any treatment is 1500 mg for children. In general, the total dose in children is 30 mg/kg. Treatment for pediatric streptococcal pharyngitis should be dosed at a different regimen (see below).

The total dose of 30 mg/kg should be given as a single daily dose of 10 mg/kg daily for 3 days, or given over 5 days with a single daily dose of 10 mg/kg on day 1, then 5 mg/kg on days 2-5. As an alternative to the above dosing, treatment for children with acute otitis media can be given as a single dose of 30 mg/kg.

For pediatric streptococcal pharyngitis, azithromycin given as a single dose of 10 mg/kg or 20 mg/kg for 3 days has been shown to be effective; however, a daily dose of 500 mg must not be exceeded. However, penicillin is the usual drug of choice in the treatment of *Streptococcus pyogenes* pharyngitis, including prophylaxis of rheumatic fever.

For children weighing less than 15 kg, azithromycin suspension should be measured as closely as possible. For children weighing 15 kg or more, azithromycin suspension should be administered according to the guide provided below:

Azithromycin Suspension 30 mg/kg Total Treatment Dose		
Weight (kg)	3-Day Regimen	5-Day Regimen
< 15	10 mg/kg once daily on days 1-3.	10 mg/kg on day 1, then 5 mg/kg once daily on days 2-5.
15-25	200 mg (5 ml) once daily on days 1-3.	200 mg (5 ml) on day 1, then 100 mg (2.5 ml) once daily on days 2-5.

26-35	300 mg (7.5 ml) once daily on days 1-3.	300 mg (7.5 ml) on day 1, then 150 mg (3.75 ml) once daily on days 2-5.
36-45	400 mg (10 ml) once daily on days 1-3.	400 mg (10 ml) on day 1, then 200 mg (5 ml) once daily on days 2-5.
> 45	Dose as per adults.	Dose as per adults.

Azithromycin tablets should only be administered to children weighing more than 45 kg.

Elderly

The same dosage as in adult patients is used in the elderly.

Elderly patients may be more susceptible to the development of torsades de pointes arrhythmia than younger patients.

Patients with Renal Impairment

No dose adjustment is necessary in patients with mild to moderate renal impairment (GFR 10 - 80 ml/min). Caution should be exercised when azithromycin is administered to patients with severe renal impairment (GFR < 10 ml/min).

Patients with Hepatic Impairment

The same dosage as in patients with normal hepatic function may be used in patients with mild to moderate hepatic impairment.

Mode of Administration

Oral

Contraindications

The use of this product is contraindicated in patients with a known hypersensitivity to azithromycin, erythromycin, any macrolide or ketolide antibiotic, or to any of the excipients.

Warnings and Precautions

Hypersensitivity

As with erythromycin and other macrolides, rare serious allergic reactions, including angioedema and anaphylaxis (rarely fatal), dermatologic reactions including Stevens-Johnson Syndrome (SJS), Toxic Epidermal Necrolysis (TEN) (rarely fatal), and Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment.

If an allergic reaction occurs, the drug should be discontinued and appropriate therapy should be instituted. Physicians should be aware that reappearance of the allergic symptoms may occur when symptomatic therapy is discontinued.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)], ZIROMIN should be discontinued immediately and appropriate treatment should be urgently initiated.

Prolongation of the QT interval

Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin.

Prescribers should consider the risk of QT prolongation, which can be fatal, when weighing the risks and benefits of azithromycin for at-risk groups including:

- Patients with congenital or documented QT prolongation
- Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones
- Patients with electrolyte disturbance, particularly in cases of hypokalemia and hypomagnesemia
- Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency
- Elderly patients: elderly patients may be more susceptible to drug-associated effects on the QT interval

Infantile hypertrophic pyloric stenosis (IHPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and/ or irritability with feeding occurs.

Since liver is the principal route of elimination for azithromycin, the use of azithromycin should be undertaken with caution in patients with significant hepatic disease. Abnormal liver function, hepatitis, cholestatic jaundice, hepatic necrosis and hepatic failure have been reported. Some of which have resulted in death. Discontinue azithromycin immediately if signs and symptoms of hepatitis occur.

In patients receiving ergot derivatives, ergotism has been precipitated by coadministration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergot and azithromycin. However, because of the theoretical possibility of ergotism, azithromycin and ergot derivatives should not be co-administered.

As with any antibiotic preparation, observation for signs of superinfection with non-susceptible organisms, including fungi is recommended.

Clostridium difficile associated diarrhea (CDAD) has been reported with use of nearly all antibacterial agents, including azithromycin, and may range in severity from mild diarrhea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C. difficile*.

C. difficile produces toxins A and B which contribute to the development of CDAD. Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

In patients with severe renal impairment (GFR <10 ml/min) a 33% increase in systemic exposure to azithromycin was observed.

Caution in diabetic patients: 5ml of reconstituted suspension contains 3685.5812 mg saccharose.

Due to the saccharose content [3685.5812 mg per 5ml], this medicinal product is not indicated for person with fructose intolerance including hereditary fructose intolerance, glucose-galactose malabsorption or saccharose-isomaltose deficiency.

Sodium content:

This medicine contains sodium. It should be considered in patients on a controlled sodium diet.

Effects on Ability to Drive and Use Machines

There is no evidence to suggest that azithromycin may have an effect on the patient's ability to drive or operate machinery.

Interactions with Other Medicaments

Antacids

No effect on overall bioavailability was seen although peak serum concentrations were reduced by approximately 25%. In patients receiving both oral azithromycin and antacids, the drugs should not be taken simultaneously.

Cetirizine

Coadministration of a 5-day regimen of azithromycin with cetirizine 20 mg at steady-state resulted in no pharmacokinetic interaction and no significant changes in the QT interval.

Didanosine (Dideoxyinosine)

Coadministration of 1200 mg/day azithromycin with 400 mg/day didanosine in HIV-positive subjects did not appear to affect the steady-state pharmacokinetics of didanosine as compared with placebo.

Digoxin

Some of the macrolide antibiotics have been reported to impair the microbial metabolism of digoxin in the gut in some patients. In patients receiving concomitant azithromycin, a related azalide antibiotic, and digoxin the possibility of raised digoxin levels should be borne in mind.

Zidovudine

Single 1000 mg doses and multiple 1200 mg or 600 mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolite. However, administration of azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells. The clinical significance of this finding is unclear, but it may be of benefit to patients.

Azithromycin does not interact significantly with the hepatic cytochrome P450 system. It is not believed to undergo the pharmacokinetic drug interactions as seen with erythromycin and other macrolides. Hepatic cytochrome P450 induction or inactivation via cytochrome-metabolite complex does not occur with azithromycin.

Ergot

Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended.

Pharmacokinetic studies have been conducted between azithromycin and the following drugs known to undergo significant cytochrome P450 mediated metabolism.

Atorvastatin

Coadministration of atorvastatin (10 mg daily) and azithromycin (500 mg daily) did not alter the plasma concentrations of atorvastatin.

Carbamazepine

No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in patients receiving concomitant azithromycin.

Cimetidine

In a pharmacokinetic study investigating the effects of a single dose of cimetidine, given 2 hours before azithromycin, on the pharmacokinetics of azithromycin, no alteration of azithromycin pharmacokinetics was seen.

Coumarin-Type Oral Anticoagulants

There have been reports of potentiated anticoagulation subsequent to coadministration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established,

consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin

Caution should be exercised before considering concurrent administration of cyclosporin and azithromycin. If coadministration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz

Coadministration of a 600 mg single dose of azithromycin and 400 mg efavirenz daily for 7 days did not result in any clinically significant pharmacokinetic interactions.

Fluconazole

Coadministration of a single dose of 1200 mg azithromycin did not alter the pharmacokinetics of a single dose of 800 mg fluconazole. Total exposure and half-life of azithromycin were unchanged by the coadministration of fluconazole, however a clinically insignificant decrease in C_{max} (18%) of azithromycin was observed.

Indinavir

Coadministration of a single dose of 1200 mg azithromycin had no statistically significant effect on the pharmacokinetics of indinavir administered as 800 mg three times daily for 5 days.

Methylprednisolone

Azithromycin had no significant effect on the pharmacokinetics of methylprednisolone.

Midazolam

Coadministration of azithromycin 500 mg/day for 3 days did not cause clinically significant changes in the pharmacokinetics and pharmacodynamics of a single 15 mg dose of midazolam.

Nelfinavir

Coadministration of azithromycin (1200 mg) and nelfinavir at steady state (750 mg three times daily) resulted in increased azithromycin concentrations. No clinically significant adverse effects were observed and no dose adjustment is required.

Rifabutin

Coadministration of azithromycin and rifabutin did not affect the serum concentrations of either drug. Neutropenia was observed in subjects receiving concomitant treatment of azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil

There was no evidence of an effect of azithromycin (500mg daily for 3 days) on the AUC and C_{max}, of sildenafil or its major circulating metabolite.

Terfenadine

Pharmacokinetic studies have reported no evidence of an interaction between azithromycin and terfenadine. There have been rare cases reported where the possibility of such an interaction could not be entirely excluded; however, there was no specific evidence that such an interaction had occurred.

Theophylline

There is no evidence of a clinically significant pharmacokinetic interaction when azithromycin and theophylline are co-administered to healthy volunteers.

Triazolam

Coadministration of azithromycin with 0.125 mg triazolam had no significant effect on any of the pharmacokinetic variables for triazolam compared to triazolam and placebo.

Trimethoprim/sulfamethoxazole

Coadministration of trimethoprim/sulfamethoxazole DS (160 mg/800 mg) for 7 days with azithromycin 1200 mg on Day 7 had no significant effect on peak concentrations, total exposure or urinary excretion of either trimethoprim or sulfamethoxazole. Azithromycin serum concentrations were similar to those seen in other studies.

Statement on Usage during Pregnancy and Lactation

Pregnancy

Animal reproduction studies have been performed at doses up to moderately maternally toxic dose concentrations. In these studies, no evidence of harm to the fetus due to azithromycin was found. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, azithromycin should be used during pregnancy only if clearly needed.

Lactation

Limited information available from published literature indicates that azithromycin is present in human milk. No serious adverse effects of azithromycin on the breast-fed infants were observed. A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from azithromycin therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

Adverse Effects / Undesirable Effects

Azithromycin is well tolerated with a low incidence of side effects.

Blood and Lymphatic System Disorders

Transient episodes of mild neutropenia have occasionally been observed although a causal relationship to azithromycin has not been established.

Ear and Labyrinth Disorders

Hearing impairment (including hearing loss, deafness and/or tinnitus) has been reported in some patients receiving azithromycin. In those cases where follow-up information was available the majority of these events were reversible.

Gastrointestinal Disorders

Nausea, vomiting, diarrhea, loose stools, abdominal discomfort (pain/cramps), and flatulence.

Hepatobiliary Disorders

Abnormal liver function.

Skin and Subcutaneous Tissue Disorders

Allergic reactions including rash and angioedema.

Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

General Disorders and Administration Site Conditions

Local pain and inflammation at the site of infusion.

In post-marketing experience, the following additional undesirable effects have been reported:

Infections and Infestations

Moniliasis and vaginitis.

Blood and Lymphatic System Disorders

Thrombocytopenia.

Immune System Disorders

Anaphylaxis (rarely fatal)

Metabolism and Nutrition Disorders

Anorexia.

Psychiatric Disorders

Aggressive reaction, nervousness, agitation, and anxiety.

Nervous System Disorders

Dizziness, convulsions (as seen with other macrolides), headache, hyperactivity, hypoesthesia, paresthesia, somnolence, and syncope. There have been rare reports of taste/smell perversion and/or loss. However, a causal relationship has not been established.

Ear and Labyrinth Disorders

Vertigo.

Cardiac Disorders

Palpitations and arrhythmias including ventricular tachycardia have been reported. There have been rare reports of QT prolongation and torsades de pointes (see **Special Warnings and Precautions for Use**).

Skin and Subcutaneous Tissue Disorders

Allergic reactions including pruritus, rash, photosensitivity, edema, urticaria, and angioedema. Rarely, serious cutaneous adverse reactions including erythema multiforme, SJS, TEN and DRESS have been reported.

Vascular Disorders

Hypotension

Gastrointestinal Disorders

Infantile hypertrophic pyloric stenosis
vomiting/diarrhea (rarely resulting in dehydration) dyspepsia, constipation, pseudomembranous colitis, pancreatitis, and rare reports of tongue discoloration.

Hepatobiliary Disorders

Hepatitis and cholestatic jaundice have been reported, as well as rare cases of hepatic necrosis and hepatic failure, which have rarely resulted in death. However, a causal relationship has not been established.

Musculoskeletal and Connective Tissue Disorders

Arthralgia.

Renal and Urinary Disorders

Interstitial nephritis and acute renal failure.

General Disorders and Administration Site Conditions

Asthenia has been reported, although a causal relationship has not been established; fatigue, and malaise.

Overdose and Treatment

Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. The typical symptoms of an overdose with macrolide antibiotics include reversible loss of hearing, severe nausea, vomiting and diarrhea. In the event of overdose, the administration of medicinal charcoal and general symptomatic treatment and supportive measures are indicated as required.

Instructions for Use and Handling, and Disposal

Ziromin powder for oral suspension can be taken with or without food.

Tap the bottle to loosen the powder. Then add 15ml of boiled and cooled water to the content of the bottle and shake well to mix uniformly. Shake immediately prior to use.

For children weighing less than 15kg, the suspension should be measured as closely as possible. For children weighing 15kg or more, the suspension should be administered using an appropriate measuring device

Storage Conditions

Before reconstitution: Store below 30°C.

After reconstitution: The suspension should be stored below 30°C. Reconstituted suspension should be used within 5 days. Discard the reconstituted suspension if more than 5 days.

Dosage Forms and Packaging Available

ZIROMIN 200 mg/5 ml Powder for Oral Suspension is packaged in 60 ml natural HDPE bottle closed with polypropylene cap with white colored induction foil, 1 dose graduated syringe (5 ml), in cardboard box with 1 patient information leaflet.

Manufacturer

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